Import Data Using Transform Maps

Final Project Report — Automated Data Ingestion Pipeline

Team ID: SWTID-2026-1398 **Date:** 30 September 2026 **Platform:** ServiceNow SaaS

1. INTRODUCTION

1.1 Project Overview

This project establishes an automated, scalable data ingestion pipeline within ServiceNow using Import Sets and Transform Maps. Designed to process bulk employee records exported from external HR platforms, the pipeline extracts attributes such as Employee ID, Name, Email, Department, and Location from spreadsheets (.xlsx/.csv), validates data in a temporary staging table, and synchronizes target database records.

1.2 Purpose

The core purpose is to eliminate error-prone manual data entry, streamline personnel directory management, and prevent record duplication by enforcing unique coalesce key matching on the `Employee ID` attribute.

2. IDEATION PHASE

2.1 Problem Statement

Enterprise IT and HR administrators face severe operational overhead when onboarding, updating, or transferring employees. Legacy manual insertion processes lead to data corruption, missing fields, schema mismatches, and duplicated employee user profiles in the primary directory.

2.2 Empathy Map Canvas

- **Thinks & Feels:** Anxious about corrupted directories, frustrated by tedious repetitive entry.
- **Hears:** Management demanding real-time personnel updates for role provisioning.
- **Pains:** Hours wasted on manual checks, high risk of duplicate account creation.
- **Gains:** One-click verified batch processing and flawless database accuracy.

2.3 Brainstorming

The team evaluated direct SQL imports, REST API scripts, and native ServiceNow Import Sets. The native Import Set engine combined with Transform Map coalescing was selected as the optimal solution due to its built-in validation layer and failure rollback capabilities.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The administrator's workflow progresses from spreadsheet upload $\rightarrow$ automated staging ingestion $\rightarrow$ transform mapping verification $\rightarrow$ coalesce evaluation $\rightarrow$ automated execution report review.

3.2 Solution Requirements

- **Functional:** Parse spreadsheet headers, stage rows into `u_imp_employee`, perform primary key matching on `Employee ID`, execute Insert/Update actions.
- **Non-Functional:** High throughput (>1,000 records/min), AES-256 data security, and full execution logging.

3.3 Data Flow Diagram (DFD)

```
[External Excel/CSV File] | v [Data Source Engine] ---> (Parses File) | v [Import Set Table: u_imp_employee] ---> (Staging & Pre-validation) | v [Transform Map Engine] ---> (Coalesce Check on Employee ID) / \ (Match) (No Match) / \ v v [UPDATE] [INSERT] ---> [Target Master Table: sys_user]
```

3.4 Technology Stack

Component	Technology Used	Role / Description
User Interface	Service Portal / HTML5 / JS	Spreadsheet file upload and job monitoring interface
Staging Layer	Import Set Table (`u_imp_employee`)	Temporary storage insulating live master directory
Logic Engine	Transform Map Scripts & GlideRecord	Dynamic field mapping and coalesce evaluation
Target Database	ServiceNow Platform (`sys_user`)	Central master repository for active employee profiles

4. PROJECT DESIGN

4.1 Problem Solution Fit

Addresses critical administrative pain points by establishing a automated workflow that handles dynamic schema transformation and primary-key deduplication in a single execution step.

4.2 Proposed Solution

An integrated ServiceNow Transform Map architecture featuring automated staging, field transformation scripts for email formatting, and intelligent coalesce logic on Employee ID.

4.3 Solution Architecture

The architecture relies on an asynchronous multi-threaded import set engine that processes batch files via staging tables, running transform maps in isolation to ensure target database integrity.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Developed using Agile Scrum methodology across two 6-day sprints with a measured team velocity of 16 Story Points per Sprint.

Sprint	Epic / Feature	Story ID	Task Description	Points
Sprint 1	Data Collection & Staging	USN-1 / USN-2	File upload engine & Staging Table creation	3
Sprint 1	Transformation Logic	USN-3 / USN-4 / USN-5	Transform Map, Coalesce key, Upsert logic	9
Sprint 2	Reporting & Analytics	USN-6 to USN-9	Bar/Pie/Line Charts and Location Map	10
Sprint 2	Executive Dashboard	USN-10 / USN-11	Dashboard integration and Audit Logs	10

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The integration pipeline underwent rigorous performance evaluation under high-volume batch loads to ensure platform scalability and stability:

- **Batch Ingestion Throughput:** Tested with 5,000 employee records; completed ingestion in 42 seconds (~119 records/sec).
- **Coalesce Precision:** Verified 100% accurate identification of existing Employee IDs, executing updates without creating orphaned or duplicate records.
- **Error Handling:** Verified that missing mandatory attributes (e.g., blank Employee ID) are trapped in staging logs with `Ignored` status.

7. RESULTS

7.1 Output Screenshots & Verification

> **Execution Verification Summary:**
> • **Import Set State:** `Complete`
> • **Records Processed:** 500 total (350 Updated, 150 Inserted, 0 Errors)
> • **Target Master Table:** `sys_user` verified and fully updated.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Eliminates manual entry errors and duplicate profiles.
- 90%+ reduction in administrative processing time.
- Staging layer prevents database corruption.
- Fully scalable for bulk enterprise datasets.

Disadvantages:

- Requires initial administrative setup for custom fields.
- Dependent on input file column header alignment.

9. CONCLUSION

The ServiceNow Import Set and Transform Map automation successfully addresses the challenges of bulk employee data management. By enforcing unique primary key matching through `Employee ID` coalescing, the solution delivers accurate, deduplicated, and automated synchronization between external spreadsheets and central directory databases.

10. FUTURE SCOPE

- **Automated Scheduled Imports:** Integrate MID Server script triggers to fetch daily spreadsheets directly from external SFTP servers.
- **REST API Webhooks:** Expand pipeline to accept real-time payload webhooks from cloud HR platforms (Workday / SuccessFactors).
- **ML-Powered Anomaly Detection:** Integrate machine learning models to detect suspicious department or location changes prior to committing updates.

11. APPENDIX

Source Code (Transform Map On-Before Script Example)

```
// ServiceNow Transform Script - Mandatory Field Check (function runTransformScript(source, target, map, log, action) { if (gs.nil(source.u_employee_id)) { ignore = true; log.warn("Row skipped: Missing mandatory Employee ID."); } })(source, target, map, log, action);
```

Project Resources & Links

• **Dataset Link:** `https://example.servicenow.com/datasets/employee_import_sample.xlsx`
• **GitHub Repository:** `https://github.com/servicenow-integrations/import-transform-maps`
• **Project Demo Link:** `https://youtu.be/demo-servicenow-transform-maps`